Abubakar Sadiq Shittu

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RESEARCH STATEMENT

My research focuses on preventing malicious entities from compromising widely used software by advancing the adoption of cryptographic defenses like *commit signing* among software developers. Using empirical and human-centered methods, I investigate developer behaviors, tooling hurdles, and adoption patterns to identify why security tools succeed, are abandoned, or can be improved. Beyond software supply chain defenses, my broader goal is to *identify generalizable design principles* that solve multi-decade grand challenges in making key management usable across complex cryptographic systems.

EDUCATION

Ph.D. Computer Science 01/25–Present
University of Tennessee Knoxville, United States GPA: 3.67/4.00

- *Passed Comprehensive Examination:* 10/26
- *Expected Graduation:* 05/28
- *Advisor:* Dr. Scott Ruoti

M.Sc. Advanced Computer Science 09/23–09/24
Newcastle University, Newcastle upon Tyne, United Kingdom GPA: 4.00/4.00

- *Advisor:* Prof. Thomas Gross

B.Sc. Computer Science, First Class Honours 03/18–06/21
Fountain University Osogbo, Nigeria GPA: 4.63/5.00

- *Advisor:* Dr. Mohammed Lawal

PUBLICATIONS

[1] **Abubakar Sadiq Shittu**, Clay Shubert, John Sadik, and Scott Ruoti. "Setting Up TLS Authentication Was Hell": A Usability Study of Client Certificate Authentication. **To appear in** *Proceedings of the 2026 ACM SIGSAC Conference on Computer and Communications Security (CCS '26), November 15–19, 2026, The Hague, Netherlands.* [**Acceptance Rate: TBD%**]. DOI: 10.1145/3830454.3846525. Preprint: arXiv:2604.14330.

[2] **Abubakar Sadiq Shittu**, John Sadik, and Scott Ruoti. "A Multi-Month Study of Git Commit Signing." **To appear in** *Proceedings of the 2026 ACM SIGSAC Conference on Computer and Communications Security (CCS '26), November 15–19, 2026, The Hague, Netherlands.* [**Acceptance Rate: TBD%**]. DOI: 10.1145/3830454.3846532. Preprint: arXiv:2608.29283.

Manuscripts Under Review

[3] **Abubakar Sadiq Shittu**, John Sadik, Farzin Gholamrezae, and Scott Ruoti. "Analysis of Commit Signing on GitHub." Manuscript under review, 2026. Preprint: arXiv:2604.14014.

[4] John Sadik, **Abubakar Sadiq Shittu**, and Scott Ruoti. "A Worldwide Survey of Authentication Practices and Ecosystems".

[5] John Sadik, **Abubakar Sadiq Shittu**, and Scott Ruoti. "Lab study of new password generation interfaces".

EXPERIENCE

Graduate Research Assistant Knoxville, Tennessee, United States
USER Lab, University of Tennessee 01/25–Present

- Designed and conducted mixed-methods longitudinal user studies with 22+ participants on security systems to identify usability, functionality, and key-management barriers to adoption and correct use; analyzed qualitative self-reports, behavioral data from server logs, and standardized usability metrics (ASQ and SUS) using inferential statistics.
- Plan and conduct stakeholder interviews to examine factors influencing successful deployment of systems.
- Synthesize findings across multiple studies to identify generalizable design principles for future systems.
- Designed a stratified sampling measurement study of GitHub commit signing and built a Docker/Celery pipeline using REST and GraphQL APIs to asynchronously collect millions of user and commit records in PostgreSQL for analysis.

Information Technology Helpdesk Intern Nigeria
Nigerian National Petroleum Corporation 03/19–08/19

- Delivered on-demand IT support to 100+ users, diagnosing and resolving hardware and software issues daily.

AWARDS

- **Travel Grant**, *Summer School on Real-World Crypto and Privacy, Croatia* (2024).
- **Chevening Scholarship**, *UK Government-funded* (2023; approximately $70,000 in tuition, stipend, and travel support).

PEER REVIEW

Journals

- Journal of Cybersecurity 2025, 2026

Conferences

- 10th ACM Conference on Fairness, Accountability, and Transparency (FAccT) 2027

Artifact Evaluation Committees

- 33rd ACM Conference on Computer and Communications Security (CCS) 2026
- Network and Distributed System Security Symposium (NDSS '27) 2026

SKILLS

- **Programming:** Python, R, SQL, C/C++.
- **Analysis:** pandas, NumPy, scikit-learn, Matplotlib, Seaborn, ggplot2, dplyr.
- **Engineering:** PostgreSQL, Docker, Celery, Redis, ETL, API Integration.
- **Research:** Prolific, Qualtrics, NVivo, MAXQDA, LaTeX.
- **Tools:** Git, SonarQube, Wireshark, Linux/Unix, Windows.

AFFILIATIONS

- Graduate Student Member, **IEEE Computer Society** 2025 – Present
- Professional Member, **Association for Computing Machinery** 2026 – Present
- Volunteer, **Tennessee Extension**